

HISTOLOGY

1. With regard to the *oral cavity*, which statement is *true*?

- a) Dentin is more calcified than enamel.
- b) Pulp is dense connective tissue.
- c) Enamel is replaced in the root by cementum.
- d) Odontoblasts make enamel.
- e) None of the above.

2. Regarding the *stomach*, which statement is *true*?

- a) Is able to withstand the acidic environment because it is lined by stratified squamous keratinized epithelium.
- b) Has its gastric glands best developed in the pylorus than the other regions.
- c) Forms temporary folds called plicae circulares when it is empty.
- d) Has its muscularis externa made by skeletal muscle fibres.
- e) None of the above.

3. Which one of the following marks best the *ileum* from the *duodenum* and *jejunum*?

- a) Long and broad leaf like villi.
- b) Absence of taenia coli.
- c) Peyer's patches.
- d) Plicae circulares.
- e) Well developed intestinal glands.

4. Which is *not* the *function of the liver*?

- a. Detoxification/removal of drugs and alcohol.
- b. Storage of calcium and bile.
- c. Storage of glycogen, fat soluble vitamins, iron and cholesterol.
- d. Production of plasma proteins and lipoproteins.
- e. Fetal RBC production.

5. With regard to the *pancreas*, which statement is *false*?

- a. Although it is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
- b. The myoepithelial cells of the striated ducts deliver bicarbonate ions into the primary secretion.
- c. The exocrine portion is a compound tubuloacinar gland, similar in structure to the parotid gland.
- d. Islets of Langerhans are incompletely delimited from the acini by Schwann cells.
- e. It is a lobulated retroperitoneal gland whose islet cells are compact mass of epithelial cells not cords or plates.

6. With regard to the *digestive system*, which statement is *true*?

- a) The mucosa of the esophagus is lined by nonkeratinized stratified squamous epithelium.
- b) The filiform papilla of the tongue contain taste buds highly sensitive to salty flavor.
- c) Brunner's glands empty their secretions into lacteals.
- d) Vermiform appendix is a blind-ended evagination of the ileum.
- e) Paneth cells of the large intestine secrete calciferol and intrinsic factors.

7. *With regard to the respiratory system, which statement is false?*

- a. *elastic fibers increase in amount from trachea to alveoli.*
- b. *The mucosa of the trachea is lined with ciliated pseudostratified columnar epithelium.*
- c. *Pulmonary surfactant is secreted by alveolar dust cells.*
- d. *Larynx is partly lined by nonkeratinized stratified squamous epithelium.*

e. Clara cells are found in the terminal bronchioles.

8. In the *respiratory system*, which statement is *true*?

- a) *Below the elastic cartilage of the bronchioles are found the tubular submucosal glands.*
- b) The paranasal sinuses are lined by ciliated pseudostratified epithelium with goblet cells.
- c) *The visceral pleura is lined by a layer of ciliated pseudostratified epithelium with goblet cells.*
- d) *a smoker's respiratory epithelium typically is depleted of goblet cells.*
- e) *the lamina propria at the level of the alveolar ducts contains mucous glands and brush cells.*

9. Which of the following is not a function of the kidney?

- a. Erythropoietin production
- b. Vitamin D modification
- c. Acid base balance
- d. Aldosterone production
- e. Renin production

10. In the *urinary system*, which statement is *false*?

- a) Juxtaglomerular cells are responsible for the secretion of rennin.
- b) Maintenance of acid-base balance in the blood is one of the main function of the mesangial cells.
- c) Antidiuretic hormone makes the epithelium of collecting duct permeable to water.
- d) The renal pyramid and its associated cortex is referred to as the renal lobe.
- e) The ureter is lined by transitional epithelium and it does not contain lymphatics.

11. In the *urinary system*, which statement is *true*?

- a). Distal convoluted tubule with brush borders and prominent apical canaliculi is longer and more convoluted than the proximal convoluted tubule.
- b). Calyces are lined by transitional epithelium.
- c) The lining epithelium of the female urethra is simple ciliated columnar and transitional epithelium.
- d) Podocytes are phagocytic cells and elaborate extracellular matrix that form the visceral layer of the Bowman's capsule.
- e) None of the above.

12. In the *testis*, which statement is *true*?

- a). Interstitial cells (of Leydig) with basophilic cytoplasm produce testicular fluid and they constitute the blood testis barrier.
- b). The seminiferous tubules are lined with a complex stratified epithelium.
- c). The secondary spermatocytes are the largest cells of the spermatogenic lineage.
- d). The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
- e). During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

13. In the *male reproductive system*, which statement is *false*?

- a) The compound tubuloalveolar glands of the prostate gland lie in a fibromuscular stroma.
- b) The epithelium lining the ductus epididymis is pseudostratified columnar.
- c) The bulbourethral gland is homologous to the greater vestibular gland in the female.
- d) The glans penis is the dilated distal end of corpus spongiosum.
- e) Skeletal muscle present in the ductus epididymis and ductus deferens is richly innervated by parasympathetic fibres.

14. In the *female reproductive system*, which statement is *false*?

- a) The ampulla of the uterine tube contains large mucosal folds that are extensively branched.
- b) Fornix of vagina is lined by stratified squamous epithelium.
- c) Primary oocyte completes its first meiotic division at the time of fertilization.
- d) Mammary gland is a modified apocrine sweat gland.
- e) Proliferative phase of uterus coincides with the development of ovarian follicles.

15. In the *female reproductive system*, which statement is *true*?

- a) Mucosa of the uterus with thick smooth muscle tunic is commonly called perimetrium.
- b) Germinal (surface) epithelium covering the ovary give rise to ova.
- c) The basic pH in the vaginal lumen, which prevents the growth of microorganisms, is due to the secretion of HCl by the tubuloalveolar glands of the vaginal walls.
- d) Corpus luteum is derived from theca interna and membrana granulosa of the ovarian follicle.
- e) During pregnancy, hCG stimulates the endometrium to grow extensively by hypertrophy and hyperplasia.

xx. The most common cells in dermis are:

- | | | |
|------------------------|------------------|----------------------|
| a. Keratinocytes. | b. Melanocytes. | c. Langerhans cells. |
| <u>d. Fibroblasts.</u> | e. Merkel cells. | |

II. EMBRYOLOGY

1. Concerning the development of the **urinary system**, which statement is **false**?
 - a) Urinary system begins to develop before the genital system.
 - b) Urachal cyst is a remnant of allantois.
 - c) The transitional epithelium lining the ureter is derived from mesoderm.
 - d) Unilateral renal agenesis is more common in males than females.
 - e) The kidneys attain their definitive adult position toward the end of the fetal period.

2. Concerning the development of the **urinary system**, which statement is **true**?
 - a) The podocytes of the Bowman's capsule are derived from mesoderm.
 - b) The connecting tubule is the adult derivative of the ureteric bud.
 - c) The urinary bladder becomes a pelvic organ during the 9th week of development.
 - d) The metanephros become functional just after birth.
 - e) Urine production begins shortly after birth.

3. With regard to **spermatogenesis**, which statement is false?
 - a. spermatids are transformed into spermatozoa by spermiogenesis.
 - b. The entire process of spermatogenesis takes approximately 72 hours.
 - c. Spermatogenesis is regulated by LH production by the pituitary.
 - d. Spermatogenesis begins at puberty.
 - e. Spermatogonia begin to increase in number at puberty.

4. With regard to the development of the **male reproductive system**, which statement is **false**?
 - a) Failure of urethral folds to fuse completely results in hypospadias.
 - b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
 - c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
 - d) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
 - e)

5. **Liquor folliculi**,
 - a) stimulates the myoepithelial cells of the mature follicle to rupture.
 - b) acts as a porous filter through which dissolved hormones and enzymes are prevented.
 - c) causes the endometrial glands to secrete and prepare the endometrium for implantation.
 - d) is composed of cumulus oophorus and polar bodies.
 - e) None of the above.

6. With regard to the development of the **female reproductive system**, which statement is **false**?
 - a) Round ligament of the uterus is formed from the gubernaculum.
 - b) The labia minora arise from urogenital folds.
 - c) The uterine tubes are derived from uterovaginal primordium.
 - d) The most common cause of testicular feminization syndrome is lack of androgen receptors.
 - e) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.

7. With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?
 - a) The cranial end of mesonephric duct in males may persist as paradidymis.
 - b) The epoophoron in the females correspond to the efferent ductules and duct of epididymis

- c) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- d) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.
- e) None of the above.

8. With regard to the development of the **endocrine system**, which of the following statements is **false**?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
- b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
- c) the thyroid gland is the first endocrine gland to develop in the embryo.
- d) the pituitary gland is ectodermal in origin.
- e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.